

Enterprise Security Monitoring Lab: Active Directory, Wazuh SIEM, and Threat Detection

Joseph Emmanuel and Varun Senthil Kumar

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Introduction

This project demonstrates how a Security Information and Event Management (SIEM) system can be implemented within an organization using Wazuh. The environment includes Active Directory running on a Windows Server, Windows client machines joined to the domain, and a Linux server hosting the Wazuh SIEM platform.

The project showcases the overall network topology, system configuration, Active Directory environment, and communication between devices over the network. It also demonstrates how Wazuh collects and centralizes security logs and events from different endpoints.

Using the Wazuh dashboard, we showcase the information collected from monitored systems, including authentication events, user activity, system changes, security alerts, and other endpoint events. The project also demonstrates how a SIEM can identify suspicious activity, generate alerts based on predefined security rules, correlate events from multiple systems, and assist administrators in investigating potential security incidents.

Overall, the project provides a practical demonstration of how organizations can use a SIEM solution to improve security visibility, monitor endpoints, detect threats, and support incident response within an Active Directory-based environment.

Topology

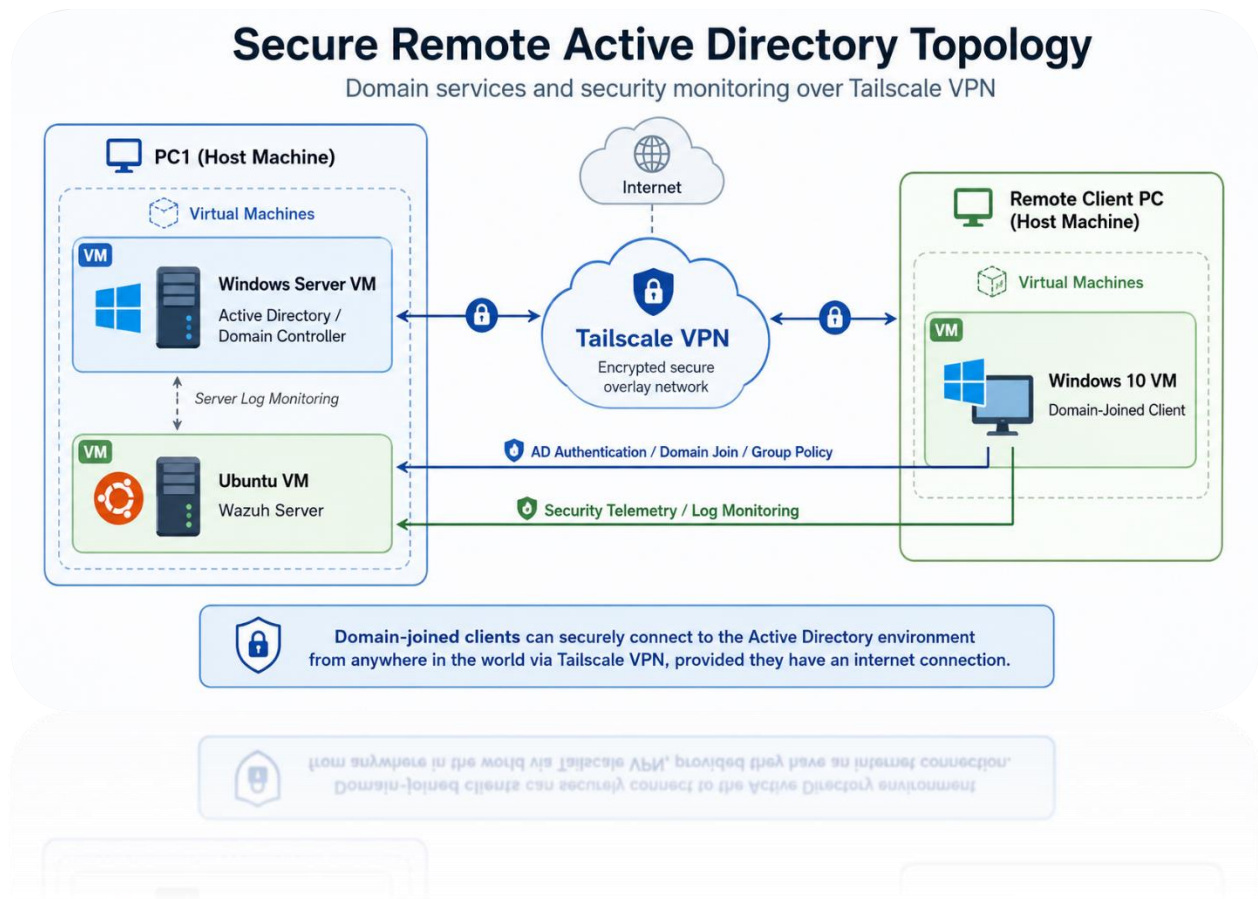


Figure 1. 1 Topology Diagram

The lab environment consists of three main virtual machines connected through a Tailscale VPN.

The Windows Server virtual machine acts as the Active Directory Domain Controller and has the Tailscale IP address **100.67.82.120**. It provides domain authentication, DNS services, user and computer management, and Group Policy for domain-joined systems.

The Ubuntu virtual machine hosts the Wazuh SIEM server and has the Tailscale IP address **100.72.72.11**. The Wazuh server receives security logs and events from monitored endpoints and provides centralized security monitoring and alerting.

The Windows 10 virtual machine acts as a domain-joined client and has the Tailscale IP address **100.104.41.126**. The client is configured to use the Active Directory Domain Controller, **100.67.82.120**, as its preferred DNS server. This allows the client to resolve the

Active Directory domain correctly and communicate with the Domain Controller for authentication, domain logon, and other Active Directory services.

All three systems are connected through Tailscale VPN, allowing secure communication between the machines regardless of their physical location, provided that they have an internet connection.

The Windows client is also monitored by Wazuh using the Wazuh agent. Security events and system information are sent from the client to the Wazuh server at **100.72.72.11**, where the information can be analyzed and displayed through the Wazuh dashboard.

```
Unknown adapter Tailscale:

Connection-specific DNS Suffix  . : taile571e8.ts.net
IPv6 Address. . . . . : fd7a:115c:a1e0::3b01:52c3
Link-local IPv6 Address . . . . . : fe80::99d0:ec2d:b2e7:536b%6
IPv4 Address. . . . . : 100.67.82.120
Subnet Mask . . . . . : 255.255.255.255
Default Gateway . . . . . :
```

Figure 1. 2 Windows Client IP Address

```
tailscale0: flags=4305<UP,POINTOPOINT,RUNNING,NOARP,MULTICAST> mtu 1280
    inet 100.72.72.11 netmask 255.255.255.255 destination 100.72.72.11
    inet6 fe80::9b9c:1e1:c271:af3d prefixlen 64 scopeid 0x20<link>
    inet6 fd7a:115c:a1e0::4f01:48cc prefixlen 128 scopeid 0x0<global>
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 500
```

Figure 1. 3 Ubuntu Machine IP Address

Active Directory

To make this lab as realistic as possible, we set up an Active Directory environment on a Windows Server virtual machine, where we created the domain **cbc.lab** to represent a fictional organization. The server was running on the IP address **100.67.82.120**.

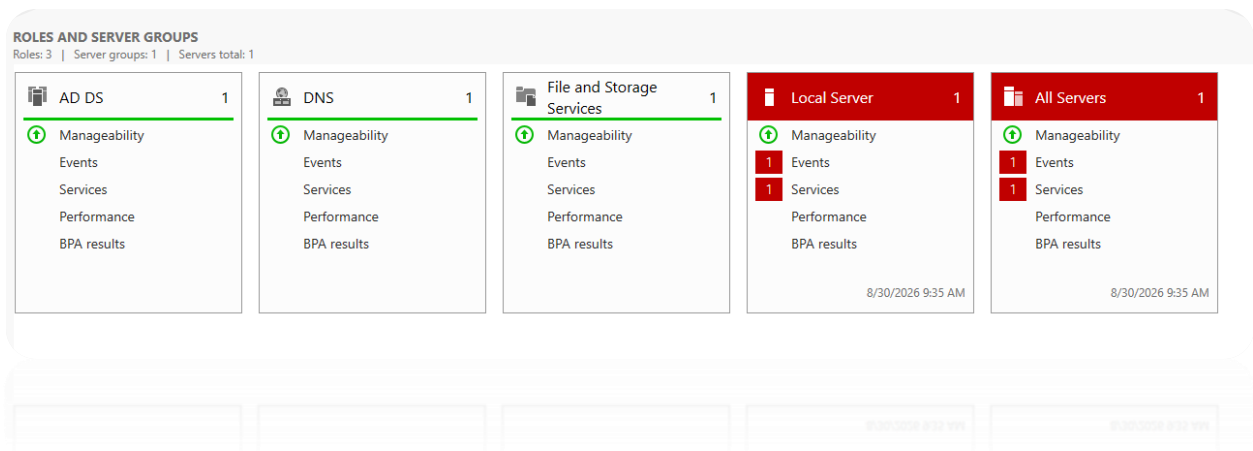


Figure 1. 4 Active Directory Roles and Servers

We created the server using the Role-based or feature-based installation option in Windows Server Manager. Since the lab environment consisted of a single Windows Server, we configured that server by installing the required roles, including Active Directory Domain Services (AD DS) and DNS. We then used Active Directory to create users, groups, and assign appropriate permissions, which will be discussed in more detail below.

```
C:\Users\Administrator>nslookup dc01.cbc.lab
Server:  magicdns.localhost-tailscale-daemon
Address:  fd7a:115c:a1e0::53

Name:    dc01.cbc.lab
Addresses: fd7a:115c:a1e0::3b01:52c3
          100.67.82.120
          192.168.197.150
```

Figure 1. 5 nslookup to the domain

The image above shows the result of a nslookup query for dc01.cbc.lab. The command successfully resolves the domain controller hostname to multiple IP addresses, confirming that DNS resolution is working correctly. The DNS server being used is reachable through the Tailscale network, indicating that Tailscale DNS configuration is being used to help resolve the domain name.

This is possible because we modified Tailscale's default MagicDNS configuration to use our domain controller as a DNS server for the cbc.lab domain.

The lookup returns two IPv4 addresses for **dc01.cbc.lab**. The first is the Tailscale IP address, **100.67.82.120**, which is the address being used for communication in this lab. The second is the local Ethernet address, **192.168.197.150**, which is not being used for communication between the lab systems.

Computer name	DC01	Last installed updates	Never
Domain	cbc.lab	Windows Update	Download updates only, using Windows Update
		Last checked for updates	Today at 9:33 AM
Microsoft Defender Firewall	Domain: On	Microsoft Defender Antivirus	Real-Time Protection: On
Remote management	Enabled	Feedback & Diagnostics	Settings
Remote Desktop	Disabled	IE Enhanced Security Configuration	On
NIC Teaming	Disabled	Time zone	(UTC-08:00) Pacific Time (US & Canada)
Ethernet0	IPv4 address assigned by DHCP, IPv6 enabled	Product ID	00455-50000-00001-AA498 (activated)
Tailscale	100.67.82.120, IPv6 enabled		
Operating system version	Microsoft Windows Server 2022 Datacenter Evaluation	Processors	AMD Ryzen 7 5800H with Radeon Graphics , AMD Ryzen 7 5800H with Radeon Graphics
Hardware information	VMware, Inc. VMware20,1	Installed memory (RAM)	7.46 GB
		Total disk space	59.37 GB

Figure 1. 6 Domain Information

The image above shows information about the Active Directory domain controller that was configured for the lab. The computer name is DC01, representing Domain Controller 1, and it is joined to the cbc.lab domain. The Server Manager window also displays information about the hardware and operating system, as well as the server's Tailscale IP address of **100.67.82.120**.

We can also see that the IPv4 address for the server's Ethernet adapter is assigned dynamically through DHCP. In a real production environment, this is generally not recommended for a domain controller because services such as Active Directory and DNS rely on the server having a consistent IP address instead, a static IP address, or a DHCP reservation, would normally be used instead. However, for this lab Tailscale provides a stable address for communication between devices, so this setup is acceptable for testing purposes.

(same as parent folder)	Start of Authority (SOA)	[68], dc01.cbc.lab., hostm...	static
(same as parent folder)	Name Server (NS)	dc01.cbc.lab.	static
dc01	Host (A)	192.168.197.150	static
dc01	Host (A)	100.67.82.120	static
dc01	IPv6 Host (AAAA)	fd7a:115c:a1e0:0000:0000:...	static
DESKTOP-VTN5UQ1	Host (A)	192.168.197.138	8/25/2026
WindowsVM	Host (A)	192.168.199.130	8/26/2026

Figure 1. 7 Domain DNS Information

The DNS records show that **dc01.cbc.lab** is configured as the authoritative name server for the cbc.lab domain. The domain controller has an A record pointing to its Tailscale IP address, **100.67.82.120**, which is the address being used for communication in this lab. Another A record, **192.168.197.150**, corresponds to the server's local Ethernet adapter, which again is not being used for communication between the lab systems.

We can also see two client hosts registered in DNS record which is the, DESKTOP-VTN5UQ1 and WindowsVM. Their current A records point to their local Ethernet IP addresses rather than their Tailscale addresses. However, for this lab, communication between the domain controller and the client systems is intended to take place over the Tailscale network. Because of this, using the clients' Tailscale IP addresses in DNS would provide a more consistent configuration for the lab environment.

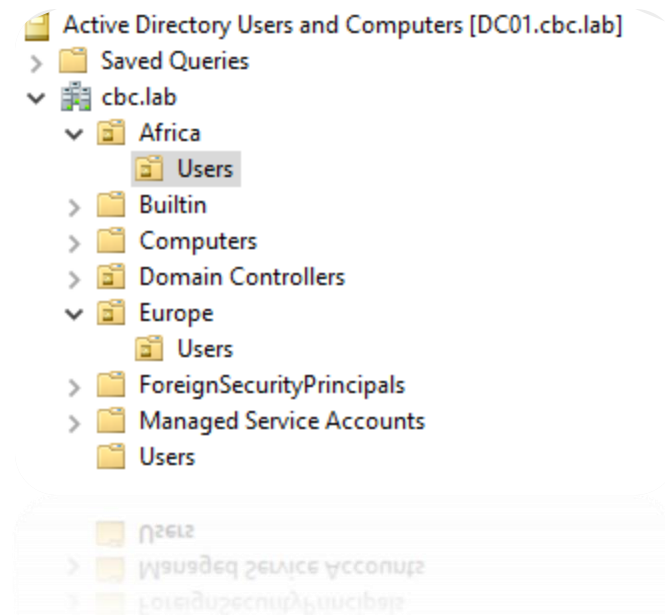


Figure 1. 8 Active Directory users and computers

The image above shows the Active Directory Users and Computers structure for the cbc.lab domain. We created separate Organizational Units (OUs) to represent different regional offices within the organization. In this example, there are OUs for Africa and Europe, and each regional OU contains a separate Users OU. This structure helps keep user accounts organized by location and makes it easier to apply different policies or permissions to users in each region.

Name	Type	Description
IT	Security Group...	
Pandi Muthu	User	
Varun Senthil	User	

Figure 1. 9 Active Directory users and groups

Within the Africa Users OU, we created a security group called IT, representing the organization's IT department, along with two user accounts, Pandi Muthu and Varun Senthil. By organizing users into groups, administrators can assign permissions and apply Group Policy settings more efficiently rather than configuring each user individually. These domain user accounts can then be used to log in to domain-joined computers using their assigned

credentials, allowing users to authenticate through Active Directory and access organizational resources based on the permissions granted to them.

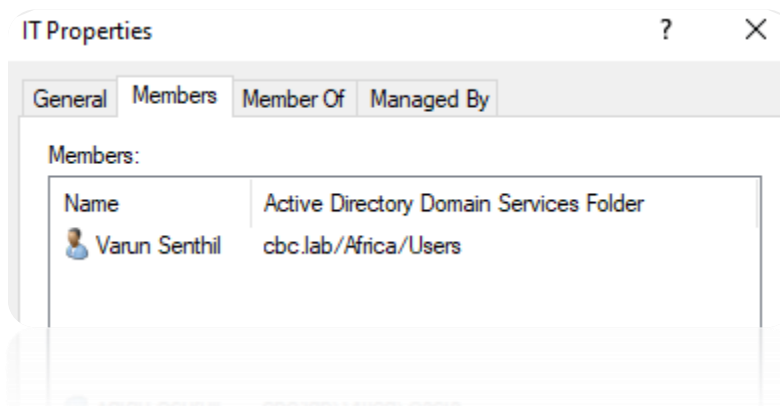


Figure 1. 10 Active Directory user

The image above shows the user Varun Senthil being added as a member of the IT security group. By adding the user to this group, Varun can inherit the permissions and access rights that have been assigned to the IT group. This makes permission management easier because administrators can grant access to the group rather than configuring permissions separately for each individual user.

Secure RDP Authentication	Enabled	None	8/30/2026 9:58:08 AM	Domain Admins (CBC\Domain Admins)
RDP Port Open	Enabled	None	8/30/2026 10:00:14 AM	Domain Admins (CBC\Domain Admins)
Enable RDP	Enabled	None	8/30/2026 10:02:02 AM	Domain Admins (CBC\Domain Admins)
Allow RDP	Enabled	None	8/30/2026 10:04:06 AM	Domain Admins (CBC\Domain Admins)

Figure 1. 11 Group Policies

We also created Group Policy settings to control Remote Desktop Protocol (RDP) access to selected computers. In this case, RDP was enabled for the computer assigned to Pandi Muthu, and members of the IT security group were granted permission to connect to that computer remotely.

- The Secure RDP Authentication policy enables Network Level Authentication (NLA). This requires users to authenticate with valid credentials before a full Remote Desktop session is established. In this domain environment, authorized domain

credentials can be validated through Active Directory, helping prevent unauthorized users from accessing the system.

- The RDP Port Open policy configures Windows Firewall to allow incoming Remote Desktop traffic, which normally uses TCP port 3389.
- The Enable RDP policy allows Remote Desktop connections on the target computer. Without this setting, the computer would not accept incoming RDP sessions even if the firewall port were open.
- The Allow RDP policy defines which users or groups are permitted to log on through Remote Desktop Services. In this case, the IT security group was granted this right, meaning members of the IT group can remotely connect to the authorized computer using their domain credentials.

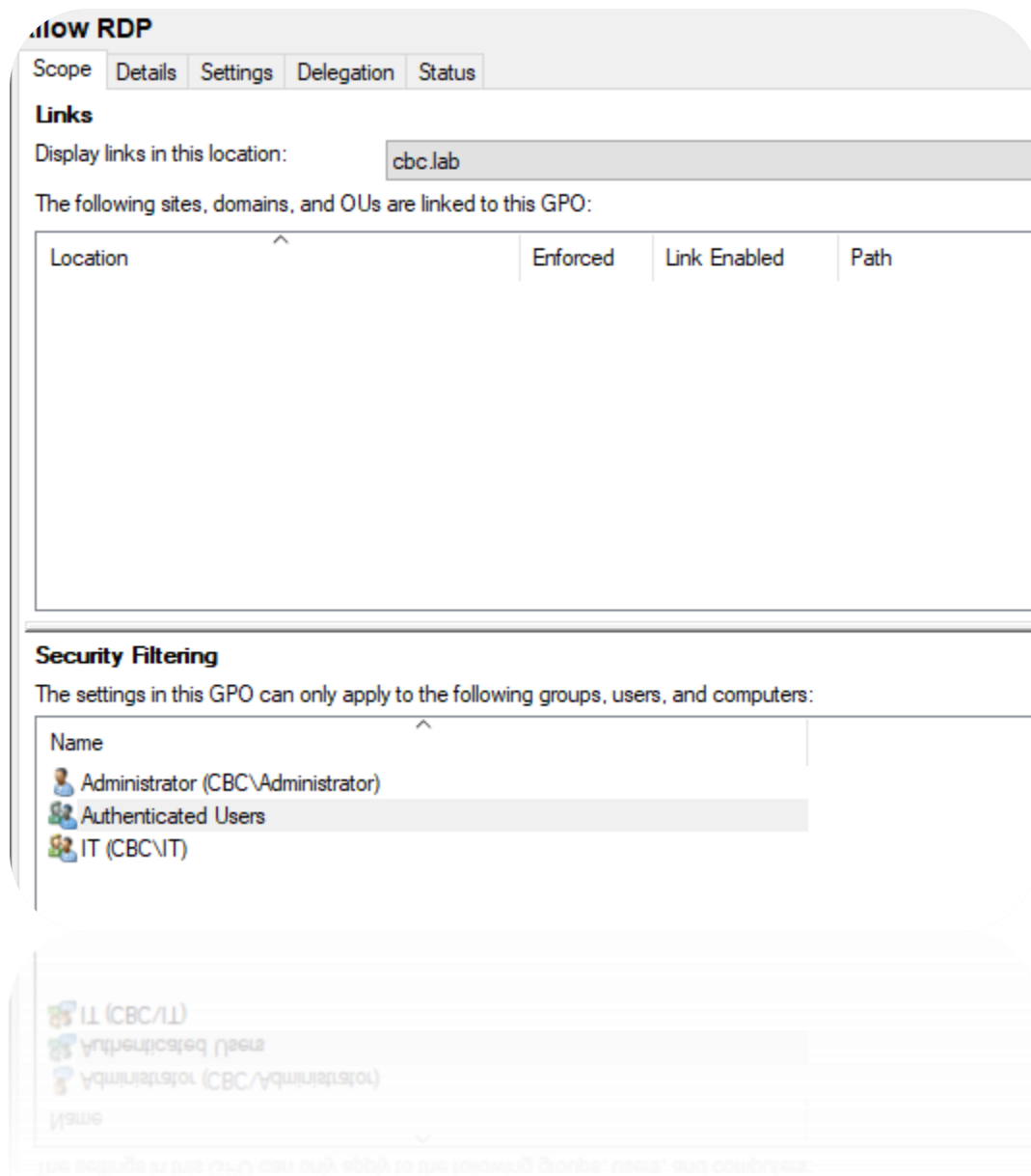


Figure 1. 12 Group Policy for RDP access

The image above shows the Security Filtering settings for the Allow RDP Group Policy Object. These settings determine which users, groups, or computers are allowed to receive this policy. In this case, the IT security group has been included, which allows members of that group to receive the RDP policy settings.

Connections

Below are images showing proof of connectivity between the machines using ping tests. We pinged the Ubuntu Server from both the Windows Server and the Windows 10 client, and the Windows Server from the Windows client. Since a successful ping requires a request to be sent and a reply to be received, the responses confirm that network communication between the tested machines is working correctly.

```
C:\Users\Administrator>ping 100.72.72.11

Pinging 100.72.72.11 with 32 bytes of data:
Reply from 100.72.72.11: bytes=32 time=18ms TTL=64
Reply from 100.72.72.11: bytes=32 time=4ms TTL=64
Reply from 100.72.72.11: bytes=32 time=6ms TTL=64
Reply from 100.72.72.11: bytes=32 time=3ms TTL=64

Ping statistics for 100.72.72.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 18ms, Average = 7ms
```

Figure 1. 13 Windows Server to Ubuntu

```
C:\Users\Administrator>ping 100.104.41.126

Pinging 100.104.41.126 with 32 bytes of data:
Reply from 100.104.41.126: bytes=32 time=14ms TTL=128
Reply from 100.104.41.126: bytes=32 time=14ms TTL=128
Reply from 100.104.41.126: bytes=32 time=13ms TTL=128
Reply from 100.104.41.126: bytes=32 time=13ms TTL=128

Ping statistics for 100.104.41.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 14ms, Average = 13ms
```

Figure 1. 14 Windows Server to Windows Client

```
C:\Users\varun>ping 100.72.72.11

Pinging 100.72.72.11 with 32 bytes of data:
Reply from 100.72.72.11: bytes=32 time=12ms TTL=64
Reply from 100.72.72.11: bytes=32 time=14ms TTL=64
Reply from 100.72.72.11: bytes=32 time=13ms TTL=64
Reply from 100.72.72.11: bytes=32 time=13ms TTL=64

Ping statistics for 100.72.72.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 14ms, Average = 13ms

C:\Users\varun>
```

Figure 1. 15 Windows Client to Ubuntu

Wazuh

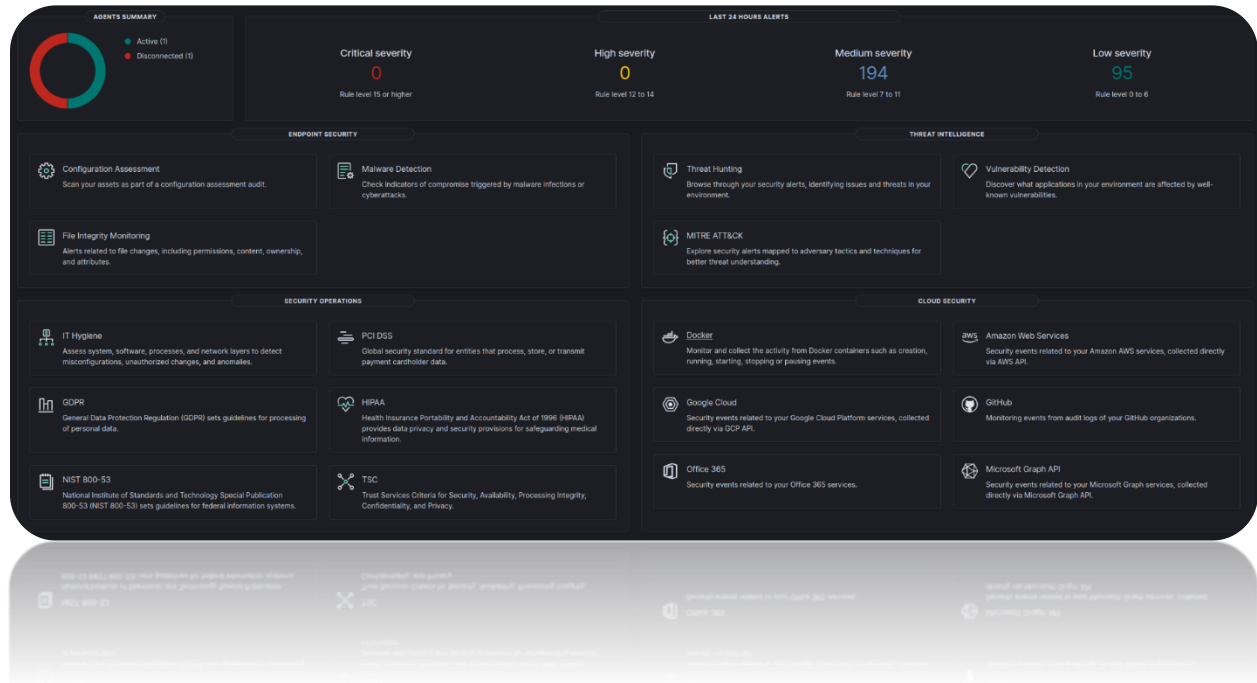


Figure 1. 16 Wazuh Dashboard

The image above shows the Wazuh dashboard and the various security modules available for monitoring and analysis. In this lab, Wazuh is installed and running on an Ubuntu Server, with a Windows Server 2022 and Windows 10 client as endpoints connected through the Wazuh agent. This report will focus on several core modules, including File Integrity Monitoring, Threat Hunting, Malware Detection, Vulnerability Detection, and Security Configuration Assessment, to demonstrate how Wazuh can be used for endpoint monitoring, threat detection, and security analysis.

We will quickly go through each module and briefly explain what it does:

- **File Integrity Monitoring (FIM)** - File Integrity Monitoring is used to detect changes made to important files and directories on monitored endpoints. Wazuh can identify when files are created, modified, or deleted and generate alerts for these events.
- **Threat Detection** - The Threat Hunting module allows you to investigate security events and search for suspicious behavior across endpoints. It provides detailed logs and alerts that can be analysed to identify possible attacks, unusual activity, or indicators of compromise
- **Malware Detection** - Malware Detection helps identify potentially malicious files or activity on endpoints. Wazuh can detect malware-related behavior using security rules, file analysis, and integration with external threat intelligence tools.
- **Vulnerability Detection** - The Vulnerability Detection module scans installed software and operating system packages on endpoints and compares them against known vulnerability databases. It can identify outdated or vulnerable software and provide information such as the CVE and severity level.
- **Security Configuration Assessment (SCA)** - Security Configuration Assessment checks endpoint configurations against predefined security policies and best-practice benchmarks. It identifies misconfigurations or insecure settings and reports whether each security check has passed or failed.

Endpoints

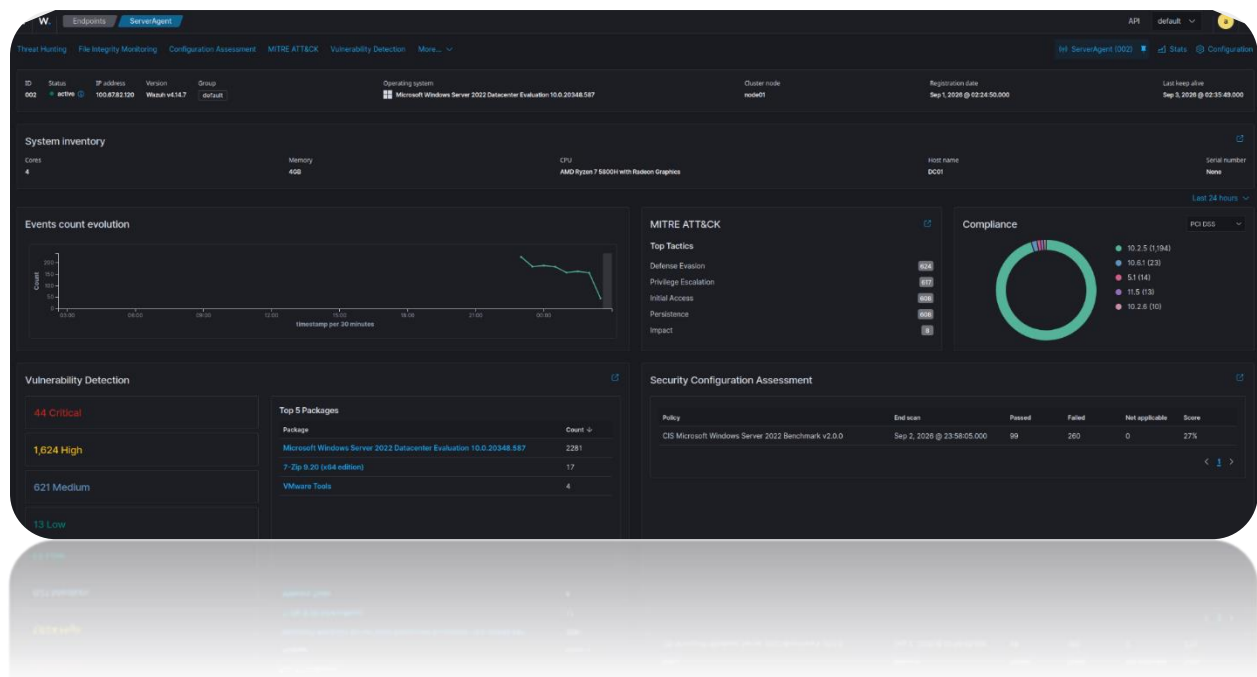
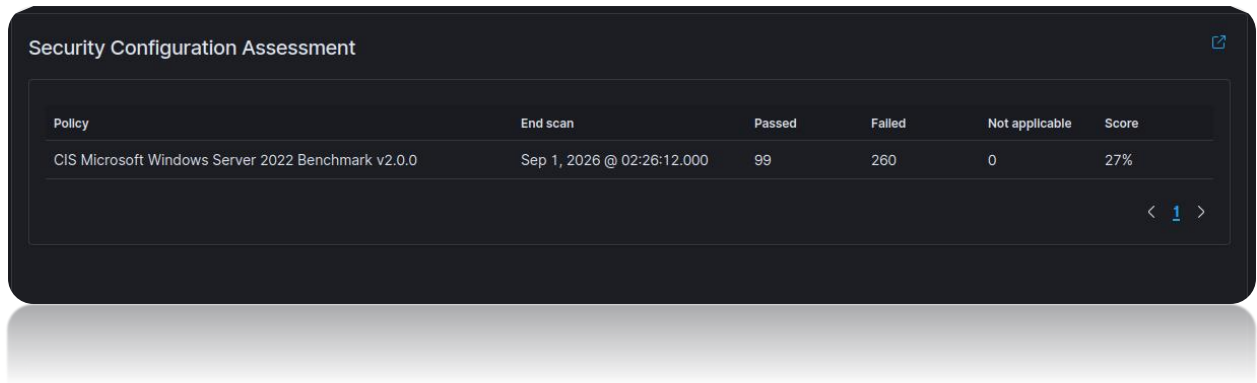


Figure 1. 17 Endpoint Dashboard

When we navigate to an individual agent endpoint, Wazuh provides a focused summary of information related to that system. This includes the operating system, Wazuh agent version, hardware details, system inventory, vulnerability information, MITRE ATT&CK mappings, and security configuration assessment results.

This view allows us to investigate a specific endpoint in much greater detail compared to the main dashboard, which provides a broader summary across all monitored systems. For example, we can quickly identify the number of critical and high-severity vulnerabilities affecting the endpoint, review security events, and see which MITRE ATT&CK tactics are associated with detected activity.

Security Configuration Assessment (SCA)



The screenshot displays a dark-themed window titled "Security Configuration Assessment". Inside, there is a table with the following data:

Policy	End scan	Passed	Failed	Not applicable	Score
CIS Microsoft Windows Server 2022 Benchmark v2.0.0	Sep 1, 2026 @ 02:26:12.000	99	260	0	27%

At the bottom right of the table, there are navigation controls: "< 1 >".

Figure 1. 18 SCA of Windows Server

The Security Configuration Assessment (SCA) module allows us to evaluate the security configuration of a system against recognized security benchmarks and best practices. These benchmarks are commonly used by organizations to measure how securely a system has been configured and to identify areas that may require improvement.

In our case, Wazuh is assessing the Windows Server 2022 system against the CIS Microsoft Windows Server 2022 Benchmark. The server passed 99 checks and failed 260 checks, resulting in an overall score of 27%.

A score of 27% indicates that there are many configuration areas that could be improved. In a real-world environment, we would review the failed checks individually, analyze the recommendations provided by Wazuh, and determine which changes are appropriate for the system. By remediating these findings, we could improve the server's overall security posture and better align it with recognized security standards.

CIS Microsoft Windows Server 2022 Benchmark v2.0.0

Passed 99 Failed 260 Not applicable 0 Score 27% End scan Sep 1, 2026 @ 02:26:12.000

Checks (359) Refresh Export formatted

Search WQL

ID ↑	Title	Target	Result
27000	Ensure 'Enforce password history' is set to '24 or more password(s)'.	Command: powershell secedit /export /c/g \$env:TEMP\secpol.cfg; Get-Content \$env:TEMP\secpol.cfg Select-String "PasswordHistorySize"; Remove-Item \$env:TEMP\secpol.cfg	Passed
27001	Ensure 'Maximum password age' is set to '365 or fewer days, but not 0'.	Command: powershell secedit /export /c/g \$env:TEMP\secpol.cfg; Get-Content \$env:TEMP\secpol.cfg Select-String "MaximumPasswordAge"; Remove-Item \$env:TEMP\secpol.cfg	Passed
27002	Ensure 'Minimum password age' is set to '1 or more day(s)'.	Command: powershell secedit /export /c/g \$env:TEMP\secpol.cfg; Get-Content \$env:TEMP\secpol.cfg Select-String "MinimumPasswordAge"; Remove-Item \$env:TEMP\secpol.cfg	Passed
27003	Ensure 'Minimum password length' is set to '14 or more character(s)'.	Command: powershell secedit /export /c/g \$env:TEMP\secpol.cfg; Get-Content \$env:TEMP\secpol.cfg Select-String "MinimumPasswordLength"; Remove-Item \$env:TEMP\secpol.cfg	Failed
27004	Ensure 'Relax minimum password length limits' is set to 'Enabled'.	Registry: HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\SAM	Failed
27005	Ensure 'Account lockout duration' is set to '15 or more minute(s)'.	Command: powershell secedit /export /c/g \$env:TEMP\secpol.cfg; Get-Content \$env:TEMP\secpol.cfg Select-String "LockoutDuration"; Remove-Item \$env:TEMP\secpol.cfg	Failed
27006	Ensure 'Account lockout threshold' is set to '5 or fewer invalid login attempt(s), but not 0'.	Command: powershell secedit /export /c/g \$env:TEMP\secpol.cfg; Get-Content \$env:TEMP\secpol.cfg Select-String "LockoutBadCount"; Remove-Item \$env:TEMP\secpol.cfg	Failed
27007	Ensure 'Reset account lockout counter after' is set to '15 or more minute(s)'.	Command: powershell secedit /export /c/g \$env:TEMP\secpol.cfg; Get-Content \$env:TEMP\secpol.cfg Select-String "ResetLockoutCount"; Remove-Item \$env:TEMP\secpol.cfg	Failed
27008	Ensure 'Accounts: Block Microsoft accounts' is set to 'Users can't add or log on with Microsoft accounts'.	Registry: HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Policies\System	Failed
27009	Ensure 'Accounts: Guest account status' is set to 'Disabled' (MS only).	Command: powershell -NoProfile -Command "[boot](Get-CimInstance -Query 'SELECT * FROM Win32_UserAccount WHERE LocalAccount = TRUE AND SID LIKE 'S-1-5-21-9-501' Where-Object -Property Disabled)"	Failed

Rows per page: 10

Figure 1. 19 Detailed SCA of Windows Server

By selecting the Security Configuration Assessment, we can view detailed results for each individual security check, including which controls passed and which failed. Wazuh also provides additional information for many checks, such as the expected configuration, the current result, and recommended remediation steps.

This allows administrators to identify misconfigurations, understand why a control failed, and apply the necessary changes to improve the overall security posture of the server and the wider environment.

For example, the assessment shows that the minimum password length requirement failed, while several password history and password age controls passed. In a domain environment, we could update the relevant Group Policy setting, which should pass the check, improving the server's security posture.

Vulnerability Detection



Figure 1. 20 Vulnerability Dashboard

The Vulnerability Detection module displays known vulnerabilities associated with the systems connected to the Wazuh server. In this lab, Wazuh identified 57 critical, 2,298 high, 845 medium, and 19 low-severity vulnerability findings across the monitored endpoints.

Many of these findings appear to be related to outdated operating system components or missing security updates. To test the module more deliberately, we installed an older version of 7-Zip on the Windows Server to see whether Wazuh could identify vulnerabilities associated with that software version.

As shown in the Top 5 packages section on the right, 7-Zip 9.20 (x64 edition) appears as one of the packages with detected vulnerabilities.



Figure 1. 21 Vulnerability Dashboard (7-zip)

After filtering the vulnerability data for the 7-Zip 9.20 package, we can see the vulnerabilities associated specifically with that version, including their corresponding CVE identifiers. These CVEs can then be researched to find remediation steps.

The detailed view shows 2,285 hits. The table below represents the first 15 rows of the results:

agent.name	package.name	package.version	vulnerability.description	vulnerability.severity	vulnerability.id
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Heap-based buffer overflow in Windows Installer allows a...	High	CVE-2020-70347
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Stack-based buffer overflow in Windows Installer allows a...	High	CVE-2020-70348
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Use after free in Windows Ancillary Function Driver for WL...	High	CVE-2020-70307
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Heap-based buffer overflow in Windows DNS allows an a...	Medium	CVE-2020-70304
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Use after free in Windows Ancillary Function Driver for WL...	High	CVE-2020-68820
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Concurrent execution using shared resource with improp...	High	CVE-2020-68802
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Relative path traversal in Windows DNS allows an authori...	Medium	CVE-2020-65785
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Heap-based buffer overflow in Windows SCSI Target Ser...	Critical	CVE-2020-65781
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Heap-based buffer overflow in Windows Message Queue...	High	CVE-2020-65780
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Use after free in Windows DNS allows an unauthorized att...	High	CVE-2020-65789
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Heap-based buffer overflow in Desktop Window Manager...	High	CVE-2020-65787
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Heap-based buffer overflow in Desktop Window Manager...	High	CVE-2020-65786
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Out-of-bounds read in Windows NTFS allows an authorize...	Medium	CVE-2020-65784
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Improper access control in Windows Kernel allows an auth...	High	CVE-2020-65773
ServerAgent	Microsoft Windows Server 2022 Datacenter Evaluation 10...	10.0.20348.587	Null pointer dereference in Windows SCSI Target Service...	High	CVE-2020-65681

Figure 1. 22 Detailed packages containing vulnerabilities

Here we can see a more detailed inventory view showing the software packages detected on the endpoint and the vulnerabilities associated with them. In this example, many of the findings are related to Windows Server operating system components, although we can navigate through the results to review other installed applications as well.

Each entry includes useful information such as the affected package, package version, vulnerability description, severity rating, and associated CVE identifier. This allows us to better understand the vulnerability and determine the appropriate remediation, such as applying a patch, updating the affected software, or removing an outdated package.

File Integrity Monitoring (FIM)

```
<!-- Monitor Downloads and Desktop for all local users -->  
  
<directories realtime="yes" check_all="yes">C:\Users\*\Downloads</directories>  
<directories realtime="yes" check_all="yes">C:\Users\*\Desktop</directories>
```

```
<!-- Monitor Downloads and Desktop for all local users -->  
  
<directories realtime="yes" check_all="yes">C:\Users\*\Downloads</directories>  
<directories realtime="yes" check_all="yes">C:\Users\*\Desktop</directories>
```

Figure 1. 23 Adding directories to checklist in the Windows wazuh config file

To set up File Integrity Monitoring (FIM), we first need to edit the Wazuh configuration file on our Windows Server and specify which directories Wazuh should monitor. For this lab, we configured Wazuh to continuously monitor the Downloads and Desktop directories.

The * wildcard in **C:\Users*\Downloads** and **C:\Users*\Desktop** allows Wazuh to monitor these directories for all local user profiles, rather than specifying a single user, and the **realtime="yes"** option enables real-time monitoring, meaning Wazuh will detect changes to files in these directories as they occur.

This configuration can also be applied centrally from the Wazuh server using an agent group configuration file. Doing so would allow the same FIM settings to be distributed to multiple Windows agents. However, for this lab, we configured the monitoring directly on the Windows Server agent.

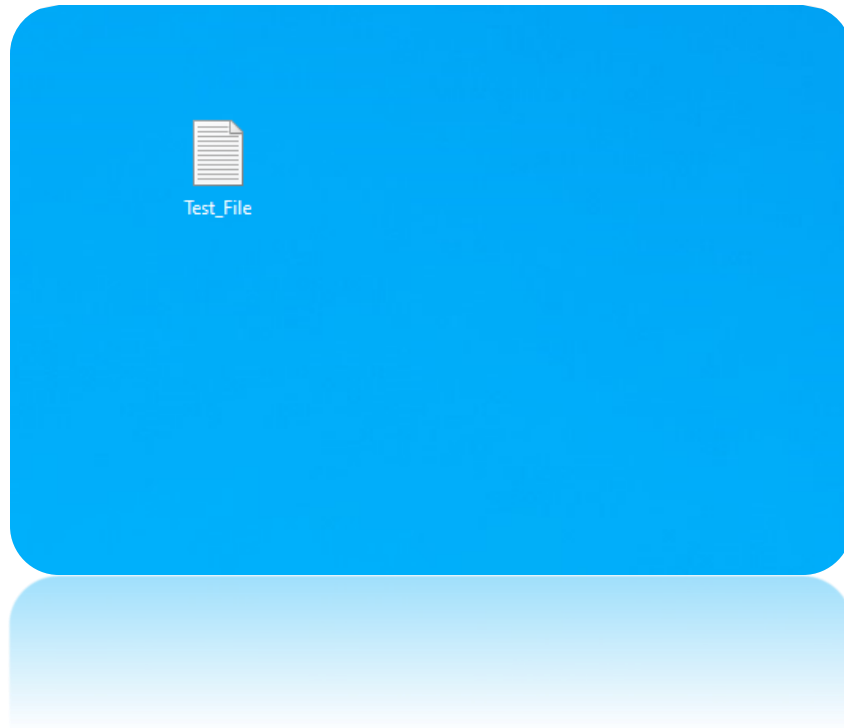


Figure 1. 24 Adding test file to desktop

To test the File Integrity Monitoring (FIM) configuration, I created a file on the Desktop called “**Test_File**” and added some data to it. Since FIM was configured to monitor changes within the Desktop and Downloads directories, modifying the file’s contents or deleting the file should generate FIM events. These events should then be forwarded to the Wazuh SIEM, where we can review the detected file changes.

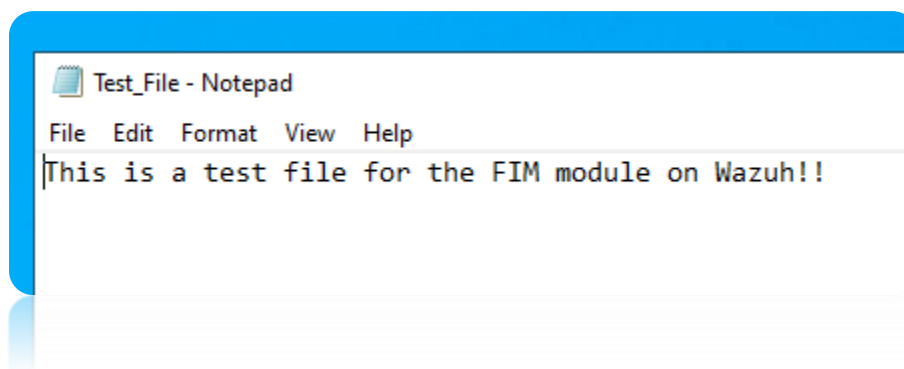


Figure 1. 25 Content inside test file

These were the contents of the test file. After creating the file and then deleting it from the Desktop, an immediate notification appeared in the SIEM under the FIM module.

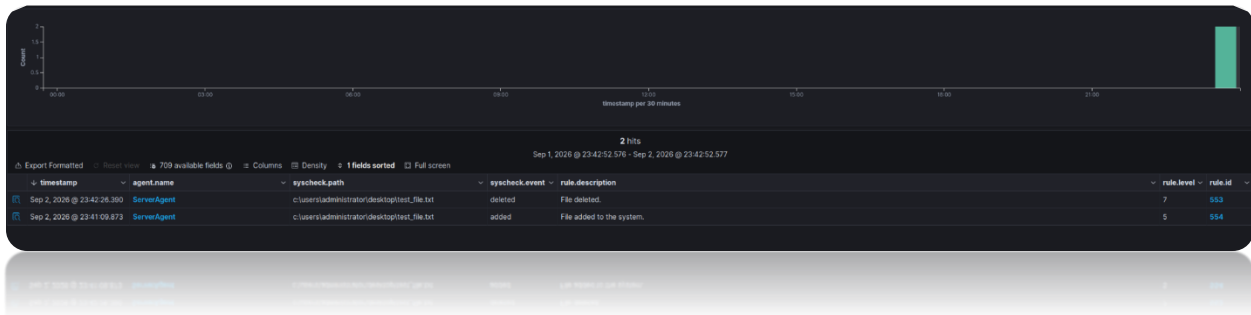


Figure 1. 26 FIM Detection

We can see two FIM events in the Wazuh dashboard. **Rule ID 554** shows that the file was added to the system, while **Rule ID 553** shows that the file was deleted. Both events were detected for the monitored Desktop path. This confirms that Wazuh successfully detected the file activity and generated the appropriate FIM alerts based on the monitoring configuration we had set up.

Malware Detection and Threat Detection

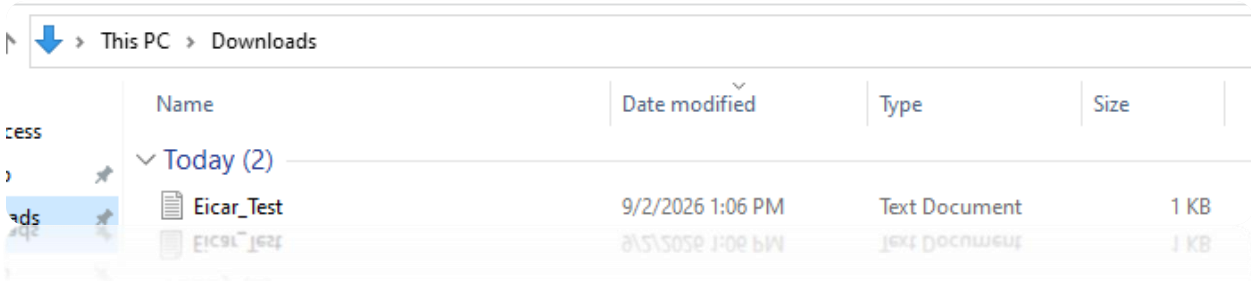


Figure 1. 27 Adding EICAR file

For the malware and threat detection test, we downloaded an EICAR test file to the Windows machine to verify whether Wazuh could detect and report the file as a potential threat.

Since the Downloads directory had already been configured for real-time File Integrity Monitoring (FIM), Wazuh should immediately detect the creation of the EICAR file within that directory. The resulting event can then be analyzed by Wazuh’s threat detection rules, allowing the SIEM manager to generate an alert if the file matches the configured malware detection criteria.



Figure 1. 28 EICAR file contents

This was the content of the EICAR test file, which should trigger an alert within the SIEM.

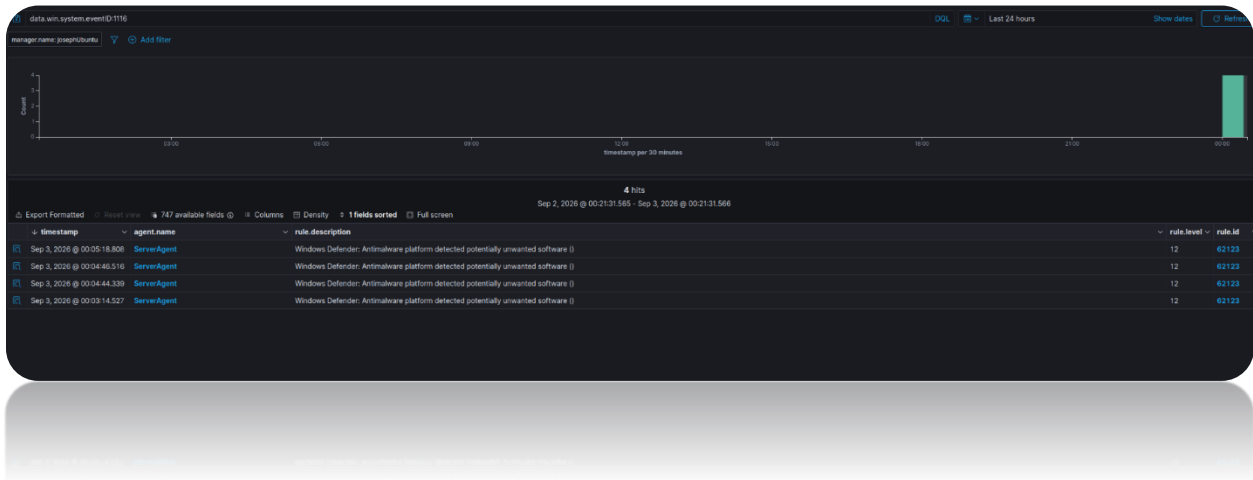


Figure 1. 29 Malware Detection (Windows Defender)

As expected, once the EICAR test file was created in the monitored Downloads directory, Windows Defender detected it as potentially unwanted or malicious software. Wazuh then collected this security event and generated a **Level 12** alert under **Rule ID 62123**. This confirmed that the SIEM was successfully receiving malware-related events from the Windows endpoint and presenting them for investigation.

Currently, Wazuh receives malware-related events generated by Windows Defender and displays them within the malware detection module. However, if we wanted to perform additional verification of suspicious files, we could integrate Wazuh with the VirusTotal API. Wazuh can use file hashes collected through File Integrity Monitoring and query VirusTotal to determine whether the file has been identified as malicious. This provides an additional layer of threat intelligence and helps us validate whether a detected file is genuinely malicious or potentially a false positive.

```
<integration>
  <name>virustotal</name>
  <api_key>[REDACTED]</api_key>
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>
```

Figure 1. 30 Adding VirusTotal API key

To integrate the VirusTotal API with Wazuh, we add the VirusTotal integration settings to the Wazuh manager configuration file on our Ubuntu server. The configuration includes the VirusTotal API key and specifies the syscheck group, which allows the integration to work with File Integrity Monitoring events.

Once the configuration has been added, we restart the Wazuh manager service so the changes take effect. After this, when FIM detects a new or modified file, Wazuh can use the file hash to query VirusTotal and check whether the file has been identified as malicious by security vendors.

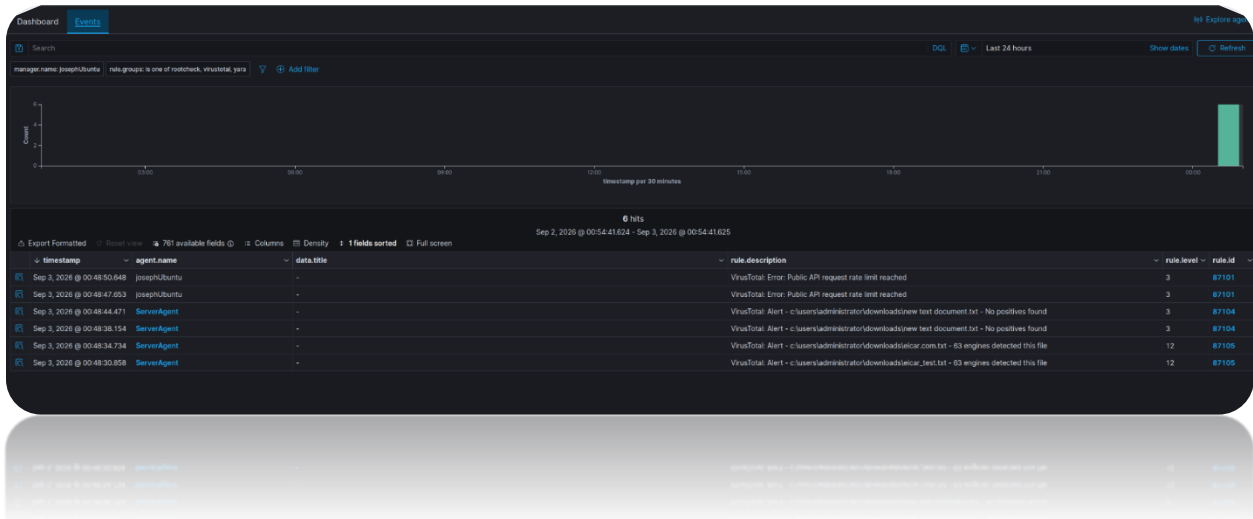
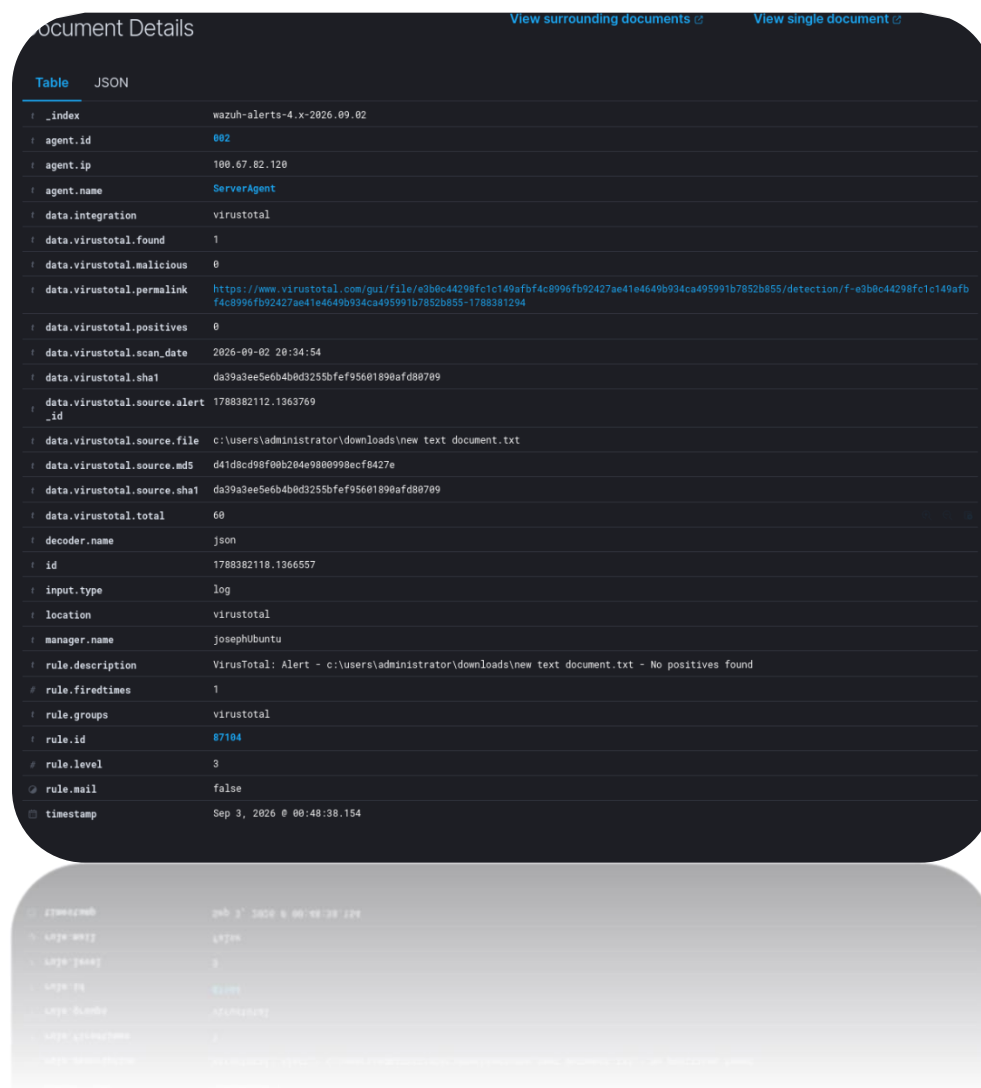


Figure 1. 31 Malware Detection (VirusTotal)

We can immediately see the VirusTotal integration returning results for both eicar_test.txt and eicar.com.txt. Although the filenames were changed while testing, the contents of the EICAR test file remained the same. VirusTotal reported that 63 security engines detected the files, which resulted in a high-severity Wazuh alert.

This is useful in a SOC environment because it provides additional threat intelligence and allows analysts to quickly see how many security vendors identify a suspicious file as malicious. This can help with alert validation and prioritization during an investigation.



The screenshot shows the 'Document Details' view in Wazuh. It features a dark-themed interface with a table of alert data. At the top, there are links for 'View surrounding documents' and 'View single document'. The table has two tabs: 'Table' (selected) and 'JSON'. The table contains the following data:

Field	Value
_index	wazuh-alerts-4.x-2026.09.02
agent.id	002
agent.ip	100.67.82.120
agent.name	ServerAgent
data.integration	virustotal
data.virustotal.found	1
data.virustotal.malicious	0
data.virustotal.permalink	https://www.virustotal.com/gui/file/e3b6c44298f1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855-1788381294
data.virustotal.positives	0
data.virustotal.scan_date	2026-09-02 20:34:54
data.virustotal.sha1	da39a3ee5e6b4b0d3255bfef95601890afd88709
data.virustotal.source.alert_id	1788382112.1363769
data.virustotal.source.file	C:\Users\Administrator\downloads\new text document.txt
data.virustotal.source.md5	d41d8cd98f00b204e9800998ecf8427e
data.virustotal.source.sha1	da39a3ee5e6b4b0d3255bfef95601890afd88709
data.virustotal.total	60
decoder.name	json
id	1788382118.1366557
input.type	log
location	virustotal
manager.name	josephubuntu
rule.description	VirusTotal: Alert - c:\users\administrator\downloads\new text document.txt - No positives found
rule.firedtimes	1
rule.groups	virustotal
rule.id	87104
rule.level	3
rule.mail	false
timestamp	Sep 3, 2026 @ 00:48:38.154

Figure 1. 32 Detailed Malware view including VirusTotal link

By opening one of the VirusTotal events in Wazuh, we can view a large amount of additional information about the file and the alert. This includes the file path, hash values such as MD5 and SHA-1, the Wazuh rule ID and severity level, the number of VirusTotal detections, and the associated agent and manager information.

The event also provides a direct link to the VirusTotal report, where we can investigate the file in more detail and see which security vendors detected it as malicious, along with their individual detection results. This additional context is useful during an investigation because it allows a SOC analyst to quickly validate the alert and gather more information about the suspicious file.

Conclusion and Future Work

This project demonstrated how Wazuh SIEM can be integrated into an Active Directory environment to provide centralized security monitoring. The lab included a Windows Server domain controller, a Windows 10 client, and an Ubuntu server hosting Wazuh, all connected through Tailscale.

Key features such as vulnerability detection, Security Configuration Assessment, File Integrity Monitoring, malware detection, and VirusTotal integration were successfully tested. Overall, the project provided practical experience with SIEM deployment, endpoint monitoring, threat detection, and basic incident investigation.

Future work could include real attack simulations using Kali Linux to evaluate how effectively Wazuh detects malicious activity, as well as creating custom Wazuh detection rules. The lab could also incorporate MITRE ATT&CK-mapped attack scenarios and automated scripts to help detect and respond to live attacks.

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VirusTotal (n.d.). *VirusTotal*. [online] www.virustotal.com. Available at: <https://www.virustotal.com/gui/> [Accessed 6 Sept. 2026].

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